

Final Project: Desktop Application with Python GUI

For the final, you will design and implement a small but **fully functional desktop GUI application** using Python. You can use everything you learned in the class. Pick either tkinter or PySide6. If you enjoyed using the Qt Designer, feel free to use it.

You will be in charge of designing the GUI. Your GUI should be as good as the GUIs you use daily. That means it should be usable and legible. Everything should fit. No text hanging off or being cut off.

Also take care of the code behind: develop a consistent naming convention and apply it to all widgets you use. No default widget names please.

Your application must demonstrate:

- GUI layout management
- Event-driven programming
- Input validation
- Data handling
- Basic user experience principles

Project Scenario

You are developing a **Student Grade Calculator**. The application should help a user input data, perform actions, and receive feedback through a graphical interface. Like this:

- Accept at least **3 numeric grades** (example: 74.5)
- Calculate and display:
 - Average
 - Format average to 2 decimal points (example: 81.25)
 - Letter grade
- Handle invalid numeric input

Core Requirements – 90 points

Your application **must include** the following:

1. GUI Design (20 pts)

- Main application window with a clear title
- At least **3 different widget types**, such as:
 - Labels
 - Entry fields
 - Buttons
- Logical and readable layout using tkinter or PySide6 layout tools

2. Event Handling (20 pts)

- Add **three buttons** with distinct actions like
- Create functions that run when the button is pressed

3. Data Processing Logic (20 pts)

- User input is captured from GUI components
- The application performs a meaningful operation, such as:
 - Calculations
 - Validation
 - Updating displayed results

4. Input Validation & Error Handling (20 pts)

- Handle invalid or missing input gracefully
- Display user-friendly error messages (e.g., using `messageBox`)
- Program must **not crash** due to bad input

5. Output & Feedback (10 pts)

- Results must be displayed within the GUI
- Clear feedback for success and failure
- Output updates dynamically based on user interaction

6. Code Quality & Design (10 points)

- Code is readable and well-organized
- Meaningful variable and function names

- Appropriate comments
- Follows Python GUI best practices

Submission Requirements

- Submit **one .py file**
- The program **must run without modification**
- Include comments at the top like this:

```
# Name:
# Course:
# Final Project:
```

Grading Rubric Summary

Category	Points
GUI Design	20
Event Handling	20
Data Logic	20
Validation & Errors	20
Output & Feedback	10
Code Quality	10
Total	100